

Biomarkers for Systemic Therapy in Metastatic Breast Cancer: ASCO Guideline Update Q and A

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In 2022, ASCO published an update of its biomarkers for systemic therapy in metastatic breast cancer (MBC) guideline.¹ The guideline update revisited recommendations from the 2015 guideline² and expanded the scope of the guideline to address:

- testing for *PIK3CA* and *ESR1* somatic variants and germline *BRCA1*, *BRCA2*, and *PALB2* pathogenic variants (mutations) to guide therapy;
- testing tumors for homologous recombination deficiency, expression of programmed cell death ligand-1 (PD-L1), deficient mismatch repair/microsatellite instability (MSI), tumor mutational burden (TMB), neurotrophic tyrosine receptor kinase (*NTRK*) gene fusions, and trophoblast cell-surface antigen 2 (TROP2) expression to determine eligibility for selected treatments; and
- use of cell-free, circulating tumor DNA (ctDNA) and circulating tumor cells for monitoring treatment response.

This companion Q and A article addresses some of the questions that clinicians may face as they implement the updated recommendations into clinical practice (Fig 1).

QUESTION: ARE ALL *PIK3CA* MUTATIONS APPROPRIATE FOR THE USE OF ALPESIB-FULVESTRANT IN PATIENTS WITH HORMONE RECEPTOR-POSITIVE METASTATIC BREAST CANCER?

Currently, treatment with alpelisib should only be considered for patients whose tumors harbor one of the 11 prespecified *PIK3CA* mutations in exons 7, 9, or 20 tested in the SOLAR-1 trial.³ The likelihood of response to alpelisib by tumors with other mutations in *PIK3CA* remains unknown.

QUESTION: WHICH TREATMENT-DIRECTING BIOMARKERS, OF THOSE THAT ARE RECOMMENDED, CAN BE DONE IN BLOOD AND WHICH REQUIRE A METASTATIC TUMOR BIOPSY AND TISSUE SUBMISSION?

PIK3CA somatic mutations can be detected in both the blood and tumor tissue, although a small proportion of mutations may be missed by the blood-based

companion ctDNA assay. We support analysis of tumor tissue for *PIK3CA* mutation assessment for any alpelisib-eligible patient who has no evidence of *PIK3CA* mutations in ctDNA.¹

We support PD-L1 testing on a metastatic tumor specimen. If fresh tumor tissue is not accessible or available, archival primary tumor tissue can be used as was done in the Keynote-355 trial. We also support testing for MSI and TMB on a metastatic tumor specimen for determination of eligibility for immunotherapy. Of note, some ctDNA assays report PD-L1, MSI, and TMB; however, this method of assessment was not used in the clinical trials examining these biomarkers, and, therefore, we do not support their use in determining eligibility for immunotherapy in patients with breast cancer. If ctDNA testing suggests the presence of one of these biomarkers, it is reasonable to send tumor tissue for confirmatory testing. *NTRK* fusions are assessed on tumor tissue, not ctDNA. For germline genetic mutation analysis, blood or nontumor tissue (usually buccal swab) is collected. If a *BRCA1/2* mutation is discovered during next-generation sequencing (NGS) analysis of tumor tissue, we support reflex germline testing.

QUESTION: WHICH BIOMARKERS ARE PRESENT IN THE PRIMARY TUMOR AND REMAIN UNCHANGED OVER THE METASTATIC COURSE OF TREATMENT AND WHICH APPEAR WITH TREATMENT?

Although *PIK3CA* mutations can emerge over the course of treatment, most are present in the primary tumor. In contrast, *ESR1* mutations emerge during the course of treatment and are generally not found in primary breast cancers. Tumor PD-L1 expression is found in both primary breast and metastatic tumor tissue, although it is higher in primary breast cancers. In addition, PD-L1 positivity in metastatic sites greatly varies by site (bone being the lowest at 12%).⁴ TMB can be found in primary breast tumors but is more common in metastatic tumor tissue.⁵ For the other biomarkers, there is not enough published information to comment. Therefore, it is important to consider the origin of the specimen when ordering assays and interpreting results.

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Biomarker Tests Recommended by the ASCO Expert Panel	Type	Evidence Quality	Strength
<i>PIK3CA</i>	EB	H	S
Germline <i>BRCA1</i> and <i>BRCA2</i>	EB	H	S
PD-L1	EB	I	S
dMMR/MSI-H	IC	L	M
TMB	IC	L	M
<i>NTRK</i> fusions	IC	L	M

Biomarker Tests Not Recommended by the ASCO Expert Panel	Type	Evidence Quality	Strength
<i>ESR1</i>	EB	Ins	M
<i>PALB2</i>	EB	L	M
HRD	IC	L	M
TROP2 expression	IC	L	M
ctDNA	IC	L	M
CTCs	IC	L	M

FIG 1. At-a-glance recommendations. CTC, circulating tumor cells; ctDNA, circulating tumor DNA; dMMR, deficient mismatch repair; EB, evidence based; H, high; HRD, homologous recombination deficiency; I, intermediate; IC, informal consensus; Ins, insufficient; L, low; M, moderate; MSI-H, microsatellite instability-high; *NTRK*, neurotrophic tyrosine receptor kinase; PD-L1, programmed death ligand 1; S, strong; TMB, tumor mutational burden.

QUESTION: WHAT IS THE OPTIMAL TIMING AND FREQUENCY OF NGS TO IDENTIFY ACTIONABLE THERAPEUTIC TARGETS IN METASTATIC BREAST CANCER?

For patients with hormone receptor–positive MBC, *PIK3CA* mutation status should be assessed while a patient is receiving first-line endocrine therapy plus CDK4/6 inhibitor to allow the use of alpelisib plus endocrine therapy at disease progression as second-line therapy. We also support early assessment of *BRCA1/2* mutational status to determine potential eligibility for PARP inhibitor therapy. Some clinically actionable genomic alterations may emerge over time under the selective pressures of various systemic therapies. For example, high TMB may be identified in a tumor sample obtained after multiple lines of systemic therapy for MBC, but not be present in the original metastatic biopsy or primary tumor. Given this, for patients with hormone receptor–positive MBC, it is reasonable to consider more comprehensive NGS testing once the tumor is clinically refractory to endocrine-based therapies.

For patients with triple-negative MBC, we support comprehensive NGS testing early in the treatment course given the more limited therapeutic options and to determine eligibility for biomarker-directed clinical trials.

Consideration can be given to repeating NGS assessment later in the metastatic treatment course to determine if any new actionable genomic targets have emerged.

QUESTION: WHAT ARE THE MOST IMPORTANT FACTORS TO COMMUNICATE TO PATIENTS REGARDING NGS AND IMPLICATIONS OF TEST RESULTS?

It is important to communicate to patients undergoing NGS testing that includes assessment of genes implicated in inherited cancer susceptibility that the identification of pathogenic germline variants is possible. This is of particular importance for patients who have not previously undergone genetic testing. Incidental identification of a pathogenic germline variant may have therapeutic implications for the patient and may also have implications for affected family members. Physicians must also be aware that pathogenic germline variants conferring increased cancer risk may not always be distinguished from tumor-specific somatic genomic alterations on NGS reports, particularly if a matched normal sample was not sequenced concurrently with tumor. If inherited cancer susceptibility is suspected due to a reported alteration in a known cancer

susceptibility gene, we support referral for genetic counseling and consideration of confirmatory testing. It is also important to communicate to patients that NGS tests may not identify any actionable findings or may have a false-negative result because of low tumor content. In

circumstances where no clinically actionable genomic alterations are identified because of no or limited tumor content, which is of particular concern with blood-based assays, we support consideration of repeat NGS testing in the future if an adequate sample can be safely obtained.

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DISCLAIMER

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EQUAL CONTRIBUTION

A.S.C., E.F.C., and Z.D. contributed equally as first authors to this work.

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